FNCE 445 Formula Sheet

Futures & Options · McDonald Ch 3–5 · Quiz 2 & Assignment 2

Annika Fajardo · Fall 2026

BA II Plus: e^x = [2nd][LN] · ln = [LN] · [STO]/[RCL] to keep full precision

Ch 3 OPTION STRATEGIES

Payoffs at expiration

```
long call  =  max(0, S_T − K)
long put   =  max(0, K − S_T)
short call = −max(0, S_T − K)
short put  = −max(0, K − S_T)
```

Profit

```
profit = payoff − FV(net premium)
FV = premium × (1 + r_period)   or   × e^(rT)
```

Put-call parity (European, same K and T)

```
C − P = PV(F₀,T − K) = (F − K)/(1 + r_period)
no-div stock:  C − P = S₀ − PV(K)
```

Price one option from the other: P = C − PV(F − K). C = P exactly when K = $F_{0,T}$.

Synthetic forward

```
long call(K) + short put(K) = long forward at K
profit after FV of premium = S_T − F₀,T
```

Parity arbitrage Compare quoted C − P with PV(F − K). Sell the expensive side: if C − P is too high, sell call + buy put (synthetic short) and buy index financed by borrowing PV(K). Profit at T = FV of today's net inflow.

Collar on a held asset (buy put K_1, sell call $K_2 > K_1$)

```
value at T: K₁ below K₁ · S_T between · K₂ above
min profit = K₁ − FV(S₀ + net prem)
max profit = K₂ − FV(S₀ + net prem)
```

Higher strike → cheaper call, pricier put. Zero-cost collar: call premium = put premium.

Asymmetric butterfly ($K_1 < K_2 < K_3$)

```
λ = (K₃ − K₂)/(K₃ − K₁)
per call written at K₂: buy λ at K₁, (1 − λ) at K₃
max payoff (at K₂) = λ(K₂ − K₁) per unit written
```

Payoff 0 below K_1 and above K_3. Break-evens: set each leg's payoff = FV(premium). Max loss = −FV(premium).

Ch 4 RISK MANAGEMENT

Natural positions Producer = long the commodity → short forward, long put (floor), or collar (long put + short call). Buyer = short → long forward, long call (cap), or short collar.

Producer profit per unit (cost = fixed + variable)

```
unhedged = S_T − cost
forward  = F − cost   (flat, short forward)
long put = S_T + max(0, K − S_T) − cost − FV(P)
collar   = S_T + max(0, Kp − S_T)
           − max(0, S_T − Kc) − cost − FV(P − C)
```

Put floor = K − cost − FV(P). Collar floor at Kp, cap at Kc. Net premium < 0 (you receive) raises both.

Produce or shut down? Fixed cost is sunk. Produce if the locked-in price > variable cost, even when total profit is negative.

Why hedging adds value Concave after-tax profit → E[f(X)] < f(E[X]), so cutting risk raises expected value. Sources: asymmetric taxes, bankruptcy/distress costs, costly external financing, debt capacity, managerial risk aversion.

```
no refund:     gain × (1 − t); loss stays full
full credit:   loss × (1 − t) → no hedging gain
carry-forward: t × loss, received a year later
```

Reasons not to hedge Transaction costs, expertise, monitoring, tax/accounting complexity, collateral.

Ch 5 FORWARDS & FUTURES

Prepaid forward

```
no dividends:      F^P = S₀
discrete divs:     F^P = S₀ − Σ Dᵢ·e^(−r·tᵢ)
dividend yield δ:  F^P = S₀·e^(−δT)
```

Forward price = FV of the prepaid

```
F₀,T = S₀·e^(rT)
F₀,T = S₀·e^((r − δ)T)
F₀,T = S₀·e^(rT) − Σ Dᵢ·e^(r(T − tᵢ))
T = days / 365
```

Value of an existing forward (delivery price K)

```
long  = (F₀,T − K)·e^(−rT) × shares
short = (K − F₀,T)·e^(−rT) × shares
no-div long per share = S₀ − K·e^(−rT)
```

Forward premium

```
F/S₀   annualized = (1/T)·ln(F/S₀)
```

Synthetics & arbitrage

```
forward = stock − zero-coupon bond
F too high → buy asset, short forward
F too low  → short asset, long forward
```

Currency forward

```
F₀,T = x₀·e^((r − r_f)T)
```

Futures vs forwards Futures: standardized, exchange-traded, clearinghouse counterparty, daily mark-to-market, initial and maintenance margin (≈70–80% of initial), easy to offset. Forwards: OTC, customized, settled at maturity.

Quiz 2 question types METHOD

TYPE	STEPS
Forward price	T = days/365 → F = S_0·e^(rT). Must be above S_0 with no dividends.
Forward value	Fair F first → (F − K) for long, (K − F) for short → × e^(−rT) → × shares.
Butterfly profit	λ → contract counts → net premium → FV at period rate → max payoff at K_2 → subtract.
Parity price	F = S_0(1 + r) → P = C − (F − K)/(1 + r).

Traps WHERE MARKS GO

- Forgetting to discount a forward's value with e^(−rT).
- Wrong sign: long gains when F > K; short gains when K > F.
- Using profit when the question asks for payoff (or the reverse).
- Skipping the opportunity cost: grow the premium to expiration.
- Mixing rate types: effective (1 + r) vs continuous e^(rT).
- Rounding rT before taking e^x. Store it.
- Collar outlay is S_0 + net premium, not just the premium.

FNCE 445 Definitions

Every position and term from Ch 3–5, in plain words

Long = you bought it / you gain if the price rises.
Short = you sold it / you gain if the price falls.

Positions WHO HAS THE RIGHT, WHO HAS THE OBLIGATION

POSITION	WHAT IT MEANS	WANTS PRICE
Long forward	Obligated to **buy** at F on the delivery date. Costs nothing today.	up
Short forward	Obligated to **sell** at F on the delivery date. Costs nothing today.	down
Long call	You paid the premium for the **right to buy** at K. Exercise only if S_T > K.	up
Short call	You received the premium and **must sell** at K if the buyer exercises.	flat / down
Long put	You paid the premium for the **right to sell** at K. Exercise only if S_T < K. Insurance.	down
Short put	You received the premium and **must buy** at K if the buyer exercises.	flat / up
Long stock	You own the asset.	up
Short stock	You borrowed and sold it; must buy it back later.	down

Buyer of an option = holder (has the right). Seller = writer (has the obligation, keeps the premium).

Option basics

Strike (K) / exercise price: the price at which the option lets you buy (call) or sell (put).

Premium: the price paid today to buy the option.

Expiration (T): the last day the option can be used. **S_T** = the asset's price that day.

European vs American: European can be exercised only at expiration; American any time before. Parity holds for European.

Moneyness: in-the-money = would pay off if exercised now (call: S > K; put: S < K). At-the-money: S = K. Out-of-the-money: wouldn't pay.

Payoff: what the position is worth at expiration, ignoring premiums.

Profit: payoff minus the future value of premiums paid (plus FV of premiums received).

Opportunity cost: the interest you give up by paying a premium today, which is why premiums are grown to expiry.

Strategies · Ch 3

Synthetic forward: long call + short put at the same K. Acts exactly like a long forward at K.

Put-call parity: $C - P = PV(F - K)$. Links call and put prices; a violation means arbitrage.

Floor: long put on an asset you own. Guarantees a minimum sale price (K).

Cap: long call on something you'll buy (or short call on what you own). Limits the price you pay (or receive).

Covered call: own the asset + sell a call. Collect premium, give up upside above K.

Collar: own the asset + buy a put (K_1) + sell a call ($K_2 > K_1$). Value is boxed between K_1 and K_2.

Zero-cost collar: a collar where the call premium received equals the put premium paid.

Bull spread: buy a low-K call, sell a high-K call. Profits if the price rises, gains capped.

Bear spread: the reverse: profits if the price falls.

Straddle: buy a call and a put at the same K. Profits from a big move either way.

Strangle: like a straddle, but the put strike is below the call strike. Cheaper, needs a bigger move.

Butterfly spread: buy calls at K_1 and K_3, write calls at K_2 in between. Pays most if S_T ends at K_2. **Asymmetric** when K_2 isn't the midpoint, so you buy λ and $(1 - \lambda)$ of the outer calls.

Risk management · Ch 4

Hedging: taking a position that offsets the risk you already have.

Natural long / short: a producer naturally gains when its output price rises (long); a buyer of an input loses (short).

Fixed vs variable cost: fixed is paid whether or not you produce (sunk); variable is paid only per unit produced.

Concave profits: a dollar lost hurts more than a dollar gained helps (taxes, distress, financing costs), so hedging raises expected value.

Forwards & futures · Ch 5

Spot price (S_0): today's price for immediate delivery.

Outright purchase: pay today, receive today. **Fully leveraged purchase:** borrow everything, receive today, repay later.

Prepaid forward (F^P): pay today, receive the asset later. Cheaper than S_0 if the asset pays dividends you'll miss.

Forward price ($F_{0,T}$): agreed today, paid and delivered at T. It's the future value of the prepaid forward.

Delivery price (K): the price locked into an existing forward. When signed, K = F, so the contract's value is 0.

Forward premium: F/S_0; annualized $(1/T)\cdot\ln(F/S_0)$.

Cost of carry: interest to hold the asset minus what it pays you: $(r - \delta)\cdot S$.

Arbitrage: a riskless profit with no net investment.

Cash-and-carry: buy the asset, short the forward. Profits when the forward is overpriced.

Reverse cash-and-carry: short the asset, long the forward. Profits when the forward is underpriced.

Futures: exchange-traded, standardized forwards with a clearinghouse and daily settlement.

Clearinghouse: becomes the counterparty to both sides, which cuts credit risk.

Notional value: contracts × multiplier × price (S&P 500: × $250).

Marking to market: gains and losses are settled into your margin account every day.

Initial margin: collateral posted to open the position. **Maintenance margin:** the minimum balance (≈70–80% of initial). **Margin call:** a demand to top back up to initial when you fall below maintenance.

Open interest: the number of contracts outstanding (buy/sell pairs).

Cross-hedge: hedging with a related but different asset; works only as well as the two are correlated.

Covered interest arbitrage: the currency version of cash-and-carry, using borrowing in one currency and lending in another.

FNCE 445 Ch 5 Playbook

Step-by-step recipes for every Ch 5 question type · Quiz 2 practice walkthroughs

First move on every question: write T in years, decide continuous or effective

Decision tree WHICH FORMULA?

QUESTION SAYS…	USE
no dividends, "forward price"	$F = S_0e^{rT}$
dividend yield δ (index)	$F = S_0e^{(r-\delta)T}$
$ dividends on dates	$F = (S_0 - \Sigma D_i e^{-rt_i}) \cdot e^{rT}$
"prepaid forward"	stop before the e^{rT}
"value of the contract", delivery price K	$(F - K)e^{-rT}$ × shares, flip for short
given F, find δ or r	$\delta = r - \ln(F/S)/T$
quoted F ≠ fair F	arbitrage: sell the expensive one
bid/ask, costs k, two rates	bounds F^+ and F^-
exchange rate	$F = x_0e^{(r - r_f)T}$
contracts, margin, price moves	mark-to-market

Recipe 1 FORWARD PRICE

1. T = days ÷ 365 (or months ÷ 12). Store it.
2. Compute rT (or $(r - \delta)T$). Store it, don't round.
3. [2nd][LN] for e^x, multiply by S_0.
4. Sanity check: no dividends → $F > S_0$. Big yield → F can be below S_0.

Recipe 2 DISCRETE DIVIDENDS

1. List each dividend with its time in years (quarterly: 0.25, 0.5, 0.75, 1).
2. PV each: $D \cdot e^{-r \cdot t}$. Add them up.
3. Prepaid $F^P = S_0 - \Sigma PV$. Forward = $F^P \cdot e^{rT}$.
4. Trap: only dividends paid **before delivery** count.

Recipe 3 VALUE OF AN EXISTING FORWARD

1. Find today's fair forward F with the remaining T.
2. Long: F − K. Short: K − F. (You win if the market moved your way.)
3. Discount: × e^{-rT}. This is the step people forget.
4. × number of shares. Check the sign once more.

Recipe 4 ARBITRAGE

1. Compute fair F. Compare with the quote.
2. **Quote too high** → cash-and-carry: borrow S_0, buy stock, short forward. Profit at T = F_quote − S_0e^{rT}.
3. **Quote too low** → reverse: short stock, lend S_0, long forward. Profit = S_0e^{rT} − F_quote.
4. Memory hook: "buy low, sell high" applies to the forward vs the synthetic (stock + loan).

Recipe 5 BOUNDS WITH TRANSACTION COSTS

$$F^+ = (S_{ask} + 2k) \cdot e^{r_{borrow} \cdot T}$$

$$F^- = (S_{bid} - 2k) \cdot e^{r_{lend} \cdot T}$$

Above F^+ cash-and-carry works (buy at ask, borrow). Below F^- reverse works (sell at bid, lend). In between: no arbitrage.

Recipe 6 IMPLIED YIELD & PREMIUM

$$\delta = r - (1/T) \cdot \ln(F/S_0)$$
$$\text{annualized premium} = (1/T) \cdot \ln(F/S_0)$$

Recipe 7 CURRENCY

$$F = x_0 \cdot e^{(r_{dom} - r_{for})T}$$

Foreign rate acts like a dividend yield. Domestic rate higher → forward above spot.

Recipe 8 FUTURES MARGIN

$$\text{notional} = \text{contracts} \times 250 \times F$$
$$\text{balance}_{new} = \text{balance}_{old} \cdot e^{r\Delta t} + \text{contracts} \times 250 \times \Delta F$$

Long gains when F rises. Below maintenance (≈70–80% of initial) → margin call back up to initial.

Quiz 2 practice walkthroughs

#	STEPS (ANSWER)
1	λ = (1,650−1,500)/(1,650−1,400) = 0.6 → buy 6 × 1,400, 4 × 1,650. Premium = 960.648 + 165.036 − 992.53 = 133.154. FV = ×1.02 = 135.817. Max payoff at 1,500 = 6 × 100 = 600. Profit = 600 − 135.817 = **464.18 (C)**
2	T = 107/365. 212.50 × e^(0.024T) = 212.50 × 1.007060 = **214.00 (C)**
3	T = 355/365. 87.25 × 1.030610 = **89.92 (B)**
4	T = 276/365. 146.80 × 1.014087 = **148.87 (D)**
5	F = 101.25e^(0.012×188/365) = 101.8777. Long: (101.8777 − 96.40) × e^(−rT) 0.993838 × 2,500 = **13,610 (B)**
6	F = 70.10e^(0.0175×0.6) = 70.8399. Short: (73.50 − 70.8399) × 0.989555 × 4,000 = **10,529 (A)**

Butterfly in 5 lines YOUR WEAK SPOT

1. $\lambda = (K_3 - K_2)/(K_3 - K_1)$ = (high − middle)/(high − low).
2. Per N written at K_2: buy λN at K_1 and (1−λ)N at K_3.
3. Net premium = bought − written. Then FV it at the period rate.
4. Max payoff happens at K_2 = λN × (K_2 − K_1). Only the K_1 calls are in the money there.
5. Max profit = max payoff − FV(premium). Max loss = −FV(premium).

Wrong answers on the quiz: "600" forgets the premium; "466.85" forgets to grow it by 2%.

FNCE 445 Your Stuck Points

From tonight's practice: the exact places things went sideways, and the fix for each

Read this the morning of the quiz.
Each box is one mistake you won't make twice.

1 · Multiply or divide?

YOU MULTIPLIED WHEN YOU SHOULD'VE DIVIDED

GOING…	EFFECTIVE RATE	CONTINUOUS RATE
today → future (FV)	× (1 + r)	× e^(+rT)
future → today (PV)	÷ (1 + r)	× e^(−rT)

Ask: is this money paid **today** (premiums) or **later** (F − K, payoffs)? Parity compares today's dollars, so (F − K) gets **divided**. Profit is measured at expiry, so premiums get **multiplied**. The minus in e^(−rT) just means "discount."

2 · Does the rate already cover the time?

"1.5% six-month rate" on a 6-month option → × 1.015 once, no T. "3% annual effective" for 6 months → × 1.03^0.5. "2.4% continuous" for 107 days → e^(0.024 × 107/365). Match the rate's period to the option's life before doing anything.

3 · S_T, payoff, profit

S_T = the asset price on expiry day. It's an unknown, like x, so answers are formulas in S_T.

POSITION	PAYOFF	PROFIT ADDS
long call	max(S_T − K, 0)	− FV(C)
short call	−max(S_T − K, 0)	+ FV(C)
long put	max(K − S_T, 0)	− FV(P)
short put	−max(K − S_T, 0)	+ FV(P)
long forward	S_T − F	nothing

Payoff ignores premiums. Profit = payoff − FV(what you paid) + FV(what you received). Call: S first. Put: K first. Short = long with a minus.

4 · The zone method

COLLARS, BUTTERFLIES, ANY COMBO

1. Put every strike on a number line. Those split S_T into zones.
2. In each zone, a call is "on" only if S_T is above its strike; a put only if below.
3. Add up only the "on" pieces. That's the piecewise payoff.
4. Test one made-up number per zone (like 1,100 or 1,400) to check. Test numbers never go in the final answer; S_T does.

Collar on stock: below K_put → you get K_put; middle → S_T; above K_call → you get K_call. Min/max profit = those caps − FV(stock + net premium).

5 · Strike vs premium

YOU GOT THE ZERO-COST COLLAR BACKWARDS

```
higher K → call CHEAPER, put PRICIER
lower K  → call PRICIER, put CHEAPER
```

Need more cash from the call you sell? Move its strike **down**, closer to today's price. Zero-cost collar with the 1,300 call short by 0.796 → strike slightly **below** 1,300.

6 · Butterfly counts

"HOW DO YOU KNOW 0.6 GOES LOW?"

```
x + y = n                    (counts cancel)
x·K₁ + y·K₃ = n·K₂           (strikes cancel)
shortcut: x = λn, λ = (K₃−K₂)/(K₃−K₁)
```

Seesaw check: the middle strike is the balance point, so **more calls go on the closer side**. Verify in 10 seconds: $\lambda K_1 + (1-\lambda)K_3$ must equal K_2. Above K_3 all the S_T terms cancel, which is why the payoff is 0 there.

7 · Arbitrage

WHICH SIDE TO SELL

Compute the fair value, compare with the quote, **sell the expensive thing, buy the cheap thing**, and hedge with the opposite synthetic. Parity violated with C − P too high (−25 vs −31.527) → sell call, buy put (sells at K), go long the forward (buys at F). Profit at T = (K − F) − FV(net premium paid). S_T always cancels; if it doesn't, a leg is missing.

8 · Value of a forward you already hold

Why F − K: to exit, you'd enter the opposite forward at today's fair F. The two cancel and you're left with F − K per share at maturity (long). Then discount it to today. Short is K − F. Locked in a better price than today's market → positive.

9 · Calculator slips

YOU GOT 2,929.26 ON PRACTICE Q6

1. Rate as a decimal: 1.75% → 0.0175.
2. T in years: 219 → 219 ÷ 365 = 0.6.
3. Exponent first: 0.0175 × 0.6 = 0.0105 [2nd][LN] = 1.01056. Store it: [STO] 1.
4. F = 70.10 × [RCL] 1. Discount factor = 1 ÷ [RCL] 1.
5. Sign check before multiplying by shares.

10 · Long question? Break it up

A butterfly max-profit question is 3 small ones: counts (λ) → max payoff at K_2 → minus FV(premium). The wrong answers are the steps you skipped: 600 = no premium, 466.85 = premium not grown. If your answer is one of those, go back one step.

Built from the FNCE 445 Ch 3–5 lecture notes (McDonald, *Fundamentals of Derivatives Markets*) and the Quiz 2 practice set. Check exam rules for what your own formula sheet may include.